Construction of Six-Panel Time-Series Figures Steady States, Transition Paths, and Optional Standard-Error Bands

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1 Objective

This note documents how to construct a six-panel monthly time-series figure overlaying data and model-implied paths for six variables: unemployment, tightness, unskilled wages, skilled wages, training share, and skilled share. The model delivers stationary equilibria for fixed regimes and an exact transition path after an unexpected permanent regime change.[1]

For each episode, the plotted model series is flat at the baseline steady state before the regime switch and then follows the transition path after the switch date. The data series may optionally be shown with standard-error bands that are computed separately within each empirical window.

2 Empirical windows and calendar mapping

The empirical design is built around two episodes, each with a baseline window and a crisis window:

Episode	Baseline window	Length	Crisis window	Length
Financial crisis	Jan 2003 – Dec 2007	60 months	Jan 2008 – Jun 2009	18 months
Covid episode	Jan 2015 – Dec 2019	60 months	Mar 2020 – Dec 2021	22 months

For each episode, construct a monthly calendar and map the regime switch as follows:

- **Financial crisis:** time index from Jan 2003 to Jun 2009, switch date in Jan 2008. Months from Jan 2003 to Dec 2007 form the pre-switch baseline segment; months from Jan 2008 to Jun 2009 form the transition segment.
- **Covid episode:** time index from Jan 2015 to Dec 2021, switch date in Mar 2020. Months from Jan 2015 to Dec 2019 form the pre-switch baseline segment; months from Mar 2020 to Dec 2021 form the transition segment.

All model and data series are defined on these calendars so that every plotted month has both a model value and a data value.

[1] See the model notes for the formal derivations of the stationary equilibria, transition laws of motion, and aggregation formulas.

3 Model objects and aggregation

The model features multiple regimes, stationary equilibria in each regime, and a transition algorithm that updates cross-sectional distributions and tightness after a regime change. For plotting purposes, the code should expose monthly series for six aggregate variables:

- u_t^{model}: aggregate unemployment rate,
- θ_t^{model}: (segment-specific or composite) tightness,
- $\bar{w}_t^{U,\text{model}}$: average unskilled wage,
- $\bar{w}_t^{S,\text{model}}$: average skilled wage,
- $TrainShare_t^{\text{model}}$: training share,
- $SkilledShare_t^{\text{model}}$: skilled share.

These series are computed from stationary equilibria and the transition path as follows.

3.1 Stationary equilibria for each regime

For each episode, solve the stationary equilibrium for:

1. the *baseline regime* z_0, using the fixed-regime equilibrium algorithm,
2. the *crisis regime* z_1, again using the fixed-regime algorithm.

From each stationary equilibrium, record:

- aggregate unemployment $u^*(z)$,
- tightness $\theta_U^*(z)$, $\theta_S^*(z)$,
- average unskilled wage $\bar{w}_U^*(z)$,
- average skilled wage $\bar{w}_S^*(z)$,
- training share $TrainShare^*(z)$,
- skilled share $SkilledShare^*(z)$.

These stationary aggregates define the flat pre-switch segment and the long-run endpoint of the transition.

3.2 Initial distribution at the switch

At the switch date T_{switch} the aggregate state moves from z_0 to z_1. The notes stipulate that value functions and policies adjust immediately to the new regime, while the cross-sectional distribution is predetermined at T_{switch} and then evolves according to the flow-balance laws of motion.

Implementation:

- Initialize the transition at time T_{switch} with the stationary cross-sectional distributions under z_0 and the new decision rules under z_1.
- Do *not* reset the distribution to the z_1 steady state at the switch.

3.3 Transition dynamics

Section 11 of the model notes provides the transition algorithm:

- Given initial distributions and policies under z_1, solve the time-dependent laws of motion for the unskilled and skilled segments forward month by month.
- At each date, update tightness using free entry and invert $q_j(\theta) = \mu_j \theta^{-\eta_j}$ to obtain $\theta_j(t)$, iterating until the tightness paths converge.

Store the transition output at monthly frequency, including the distributions and objects needed to aggregate the six variables at each t.

3.4 Aggregation of the six variables

The exact implementation depends on how the densities are discretized in the code, but the aggregation logic is as follows.

Unemployment rate. Let $U_U(t)$ denote the total unskilled unemployment mass and $U_S(t)$ the total skilled unemployment mass, and $N(t)$ the aggregate population.

$$u_t^{\text{model}} = \frac{U_U(t) + U_S(t)}{N(t)}. \tag{1}$$

Tightness. Use free-entry tightness along the transition:

$$\theta_t^{\text{model}} = \begin{cases} \theta_U^*(z_0) \text{ or } \theta_S^*(z_0), & t < T_{\text{switch}}, \\ \theta_U(t) \text{ or } \theta_S(t), & t \geq T_{\text{switch}}, \end{cases} \tag{2}$$

depending on whether unskilled, skilled, or a documented composite measure is plotted. The data definition of tightness (e.g. vacancies divided by unemployment) should be mirrored in the model construction.

Unskilled wages. The unskilled wage schedule is $w_U(x, p, z)$. Let $m_U^{\text{emp}}(x, p, t)$ be the mass of unskilled employed matches at state (x, p) in month t. Then

$$\bar{w}_t^{U,\text{model}} = \frac{\sum_{x,p} w_U(x, p, z_t)\, m_U^{\text{emp}}(x, p, t)}{\sum_{x,p} m_U^{\text{emp}}(x, p, t)}. \tag{3}$$

Skilled wages. The skilled block has two wage schedules, $w_S^0(x, p, z)$ and $w_S^1(x, p, z)$, for non-searchers and on-the-job searchers. Let $m_{S,0}^{\text{emp}}(x, p, t)$ and $m_{S,1}^{\text{emp}}(x, p, t)$ be the corresponding employment masses. Then

$$\bar{w}_t^{S,\text{model}} = \frac{\sum_{x,p} w_S^0(x, p, z_t)\, m_{S,0}^{\text{emp}}(x, p, t) + \sum_{x,p} w_S^1(x, p, z_t)\, m_{S,1}^{\text{emp}}(x, p, t)}{\sum_{x,p} m_{S,0}^{\text{emp}}(x, p, t) + \sum_{x,p} m_{S,1}^{\text{emp}}(x, p, t)}. \tag{4}$$

Training share. Let $T(t)$ be the total mass currently in training.

$$TrainShare_t^{\text{model}} = \frac{T(t)}{N(t)}. \tag{5}$$

Skilled share. Let $N_S(t)$ be the total mass with trained status (employed or unemployed) in month t.

$$SkilledShare_t^{\text{model}} = \frac{N_S(t)}{N(t)}. \tag{6}$$

3.5 Splicing steady state and transition

For each variable $X \in \{u, \theta, \bar{w}^U, \bar{w}^S, TrainShare, SkilledShare\}$, construct the model series as:

$$X_t^{\text{model}} = \begin{cases} X^*(z_0), & t < T_{\text{switch}}, \\ X_t^{\text{trans}}, & t \geq T_{\text{switch}}, \end{cases} \tag{7}$$

where $X^*(z_0)$ is the baseline steady-state aggregate and X_t^{trans} is the monthly transition path computed as above. This produces a flat pre-switch model line and a dynamic post-switch path.

4 Data series and standard errors

For each episode, prepare aligned monthly data series:

$$u_t^{\text{data}}, \quad \theta_t^{\text{data}}, \quad \bar{w}_t^{U,\text{data}}, \quad \bar{w}_t^{S,\text{data}}, \quad TrainShare_t^{\text{data}}, \quad SkilledShare_t^{\text{data}},$$
$$\text{for } t \in \{\text{episode calendar}\}.$$

Units (levels, logs, percentage points) should match the model definitions used in the plots. If necessary, transform the model series to the same units as the data before plotting.

4.1 Window-specific standard errors

Standard errors are computed separately within each empirical window:

- Baseline and crisis for the financial-crisis episode,
- Baseline and crisis for the Covid episode.

For a given variable X, define a standard-error array se_t^X that is piecewise constant within each window (or month-specific if a richer estimator is used). The data band at time t is then:

$$X_t^{\text{lower}} = X_t^{\text{data}} - c \cdot \text{se}_t^X, \tag{8}$$
$$X_t^{\text{upper}} = X_t^{\text{data}} + c \cdot \text{se}_t^X, \tag{9}$$

where $c = 1.96$ for approximate 95-percent confidence intervals or another chosen multiplier.

4.2 Optional display of bands

In the code that generates the figures, introduce a logical switch

```
SHOW_SE_BANDS = True
```

(or an equivalent option in the plotting language).

- If `SHOW_SE_BANDS = True`, plot the shaded area between X_t^{lower} and X_t^{upper} around the data line.
- If `SHOW_SE_BANDS = False`, suppress the shaded area and plot only the data line and the model line.

This switch affects only the visualisation of the data uncertainty; it does not alter the underlying estimates of standard errors.

5 Figure layout and conventions

The main deliverable is a six-panel figure per episode. A convenient layout is a 3×2 grid with panels ordered as:

1. Unemployment,
2. Tightness,
3. Unskilled wages,
4. Skilled wages,
5. Training share,
6. Skilled share.

Each panel contains:

- the monthly data series,
- the optional shaded standard-error band (data),
- the model series (steady state plus transition),
- a vertical line at T_{switch} marking the regime change.

Stylistically:

- Data: black solid line.
- Data standard-error band: light gray, semi-transparent.
- Model: blue solid or dashed line.
- Switch date: vertical dashed line in red or dark gray.

Using a common date axis across all six panels clarifies the timing of the transition in every variable.

6 Coding architecture (conceptual)

Although this note is language-agnostic, a clean implementation uses three layers:

1. **Model aggregation layer** that takes stationary and transition outputs and returns the six model series.

2. **Data preparation layer** that reads raw data, aligns it to the model calendar, and attaches window-specific standard errors.
3. **Plotting layer** that receives prepared model and data series, the switch date, and the boolean `SHOW_SE_BANDS`, and produces the six-panel figure.

This separation mirrors the economics of the problem (steady state vs. transition vs. empirical mapping) and simplifies debugging and extension.